



Swiss Institute of
Bioinformatics

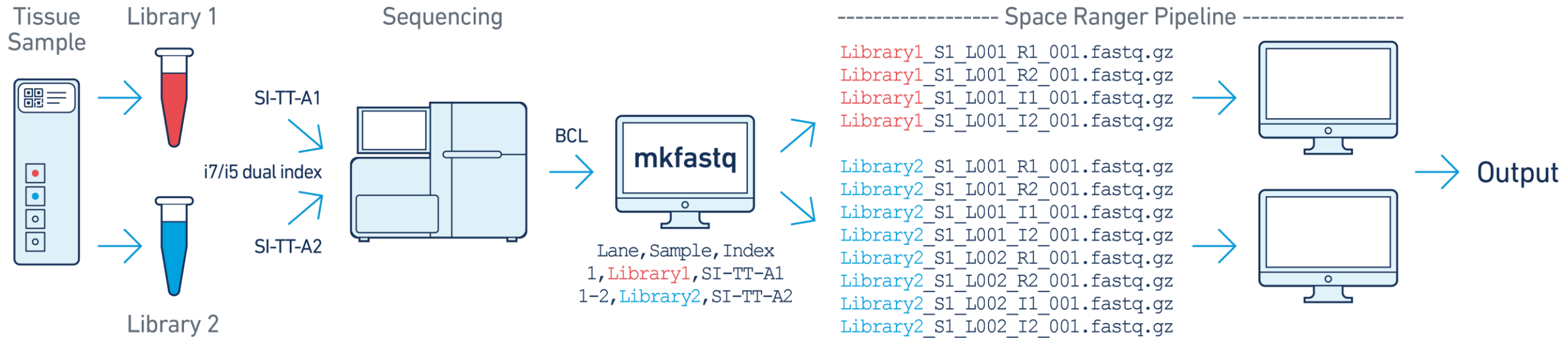
INTRODUCTION TO SPATIAL TRANSCRIPTOMICS DATA ANALYSIS

Preprocessing & Data Structures

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June 03-05, 2026

**What is the pipeline (with tools) to
obtain counts by features metrics
for single-cell transcriptomics
data?**

Workflow of obtaining read counts



What does this sample name
mean?

Library1_S1_L001_R1_001.fastq.gz

Library1_S1_L001_R1_001.fastq.gz

Component	Meaning
Library1	Library name from tissue sample
S1	Sample number
L001	Lane number
R1/R2	Read direction (forward/reverse)
I1/I2	Index reads (i7/i5)
001	Chunk number
.fastq.gz	FASTQ format, gzip compressed

Space-ranger

Key Metrics

393,543

Number of 8 μ m binned
squares under tissue

849.2

Mean reads per 8 μ m bin

226.4

Mean UMIs per 8 μ m bin

18,991

Total genes detected

Mapping

Reads Mapped to Genome	85.1%
Reads Mapped Confidently to Genome	81.6%
Reads Mapped Confidently to Intergenic Regions	5.9%
Reads Mapped Confidently to Intronic Regions	1.6%
Reads Mapped Confidently to Exonic Regions	74.1%
Reads Mapped Confidently to Transcriptome	72.7%
Reads Mapped Antisense to Gene	0.6%

Space-ranger

```

binned_output :
├─ square_002um
├─ square_008um
├─ square_016um
├─ analysis
│   ├── clustering
│   ├── diffexp
│   ├── pca
│   ├── tsne
│   └─ umap
├─ cloupe.cloupe
├─ filtered_feature_bc_matrix
│   ├── barcodes.tsv.gz
│   ├── features.tsv.gz
│   └─ matrix.mtx.gz
├─ filtered_feature_bc_matrix.h5
├─ spatial
│   └─ LV123
│       ├── scalefactors_json.json
│       └─ tissue_hires_image.png
└─ web_summary.html
```

Space-ranger: web_summary.html

Key output files

filtered_feature_bc_matrix/

Barcodes passing tissue detection

raw_feature_bc_matrix/

All barcodes (on + off tissue)

spatial/tissue_positions.csv

X/Y coordinates per barcode

Sequencing metrics

Q30 bases in RNA read

Sequencing saturation

Number of reads

Mapping metrics

Reads mapped to genome

Reads mapped confidently to transcriptome

Reads mapped to intronic regions

Spot/bin metrics

Number of spots under tissue

Median genes per spot

Median UMI counts per spot

Tissue detection QC

Tissue image + spot overlay

Clustering (UMAP/t-SNE) preview

Visium HD: binning strategy

square_002um

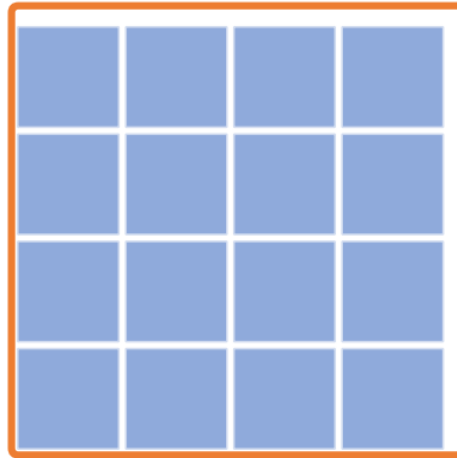
1 native capture spot



2 µm × 2 µm
highest resolution
most sparse

square_008um

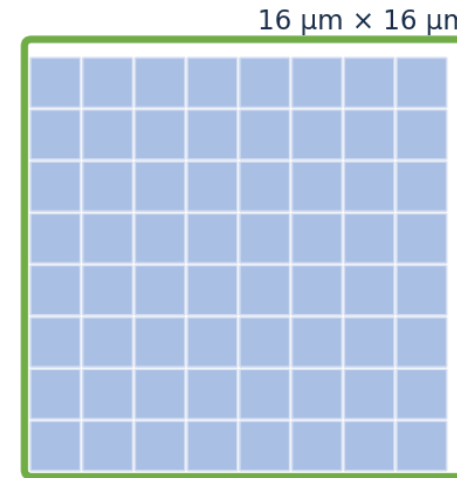
pool of 16 spots (4×4)



8 µm × 8 µm
~cell-scale
balanced depth

square_016um

pool of 64 spots (8×8)



16 µm × 16 µm
multi-cell, deeper counts
used in this course ✓

Which bin size should you use?

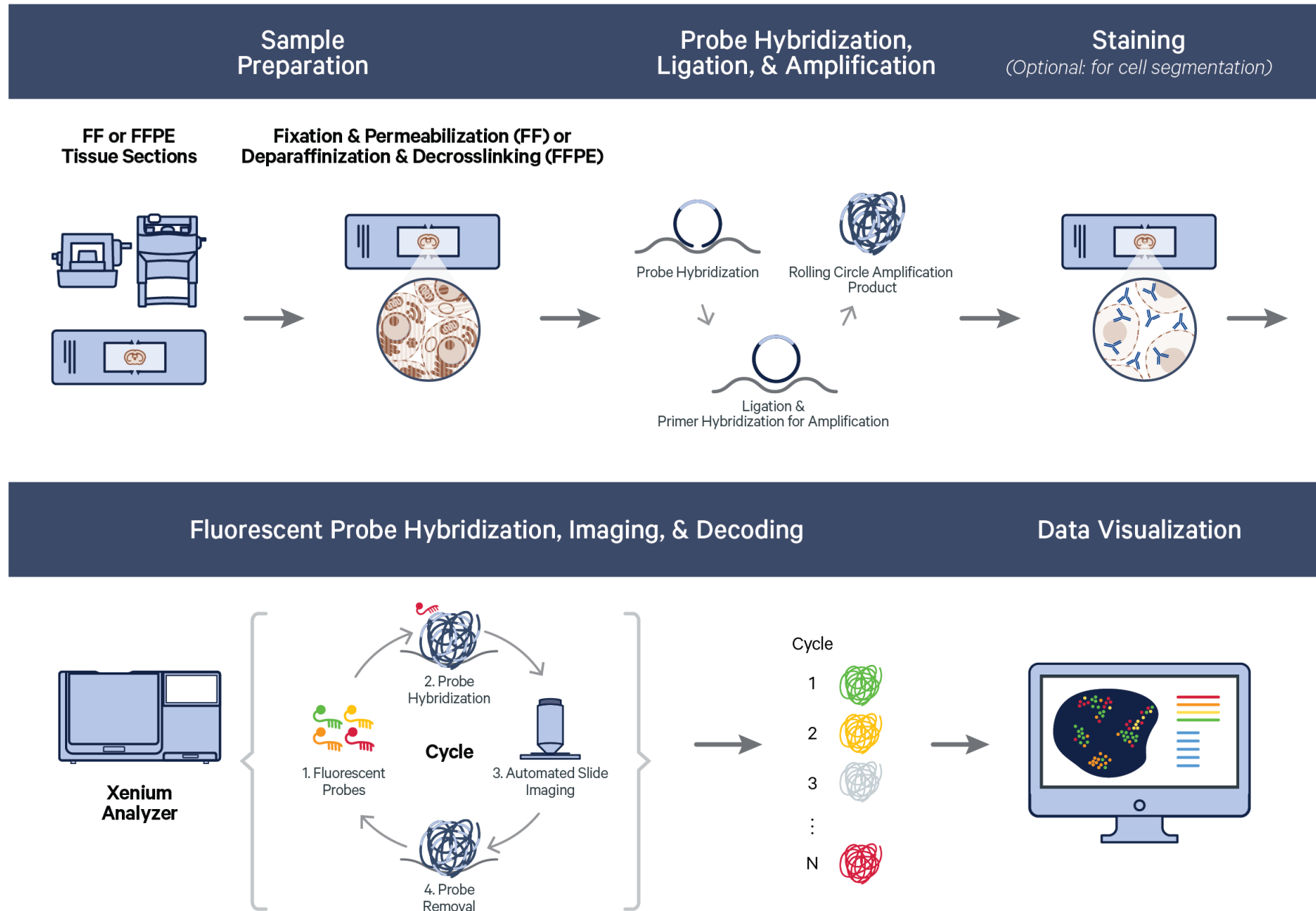
2 μm – sub-cellular resolution, very sparse counts. Use only if sub-cellular structure matters.

8 μm – roughly cell-scale, moderate depth. Good for detailed analysis.

16 μm – multi-cell bins, densest counts. Best for first-pass, QC, and this course.

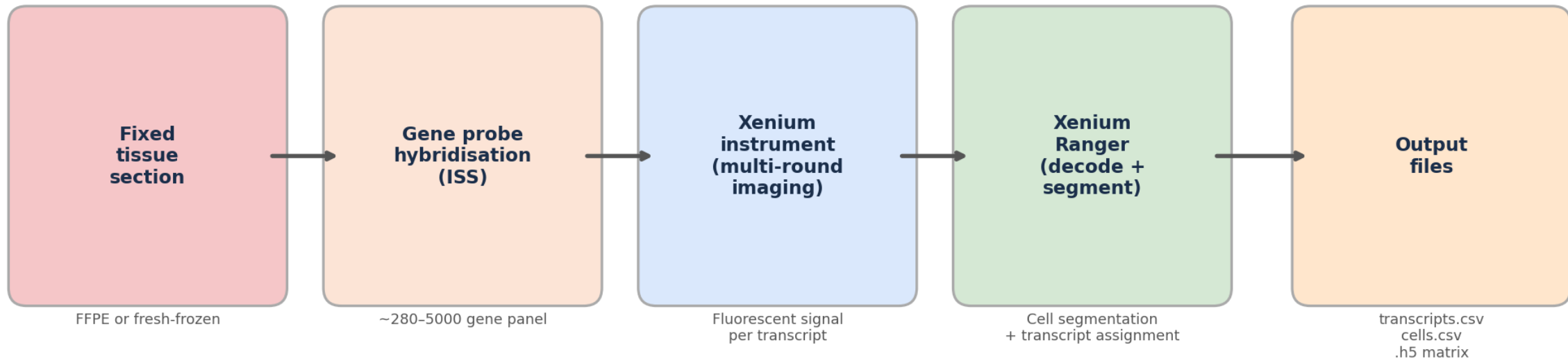
→ In this course we use `bin_size = "016"` for speed and count depth.

Xenium: pipeline overview



Xenium: pipeline overview

Imaging-based — no sequencing required



Xenium: output files

Key difference vs. Visium HD

No spot grid — coordinates are in **physical microns**

No spatial/ folder or scalefactors

Alignment tool: **Xenium Ranger / Explorer**

transcripts.csv.gz

One row per detected transcript
x_location, y_location (microns)

cells.csv.gz

Cell metadata: centroid, area, QC metrics

cell_feature_matrix.h5

Cells x genes count matrix (HDF5)

nucleus_boundaries.csv.gz

Polygon vertices per nucleus

morphology.ome.tif

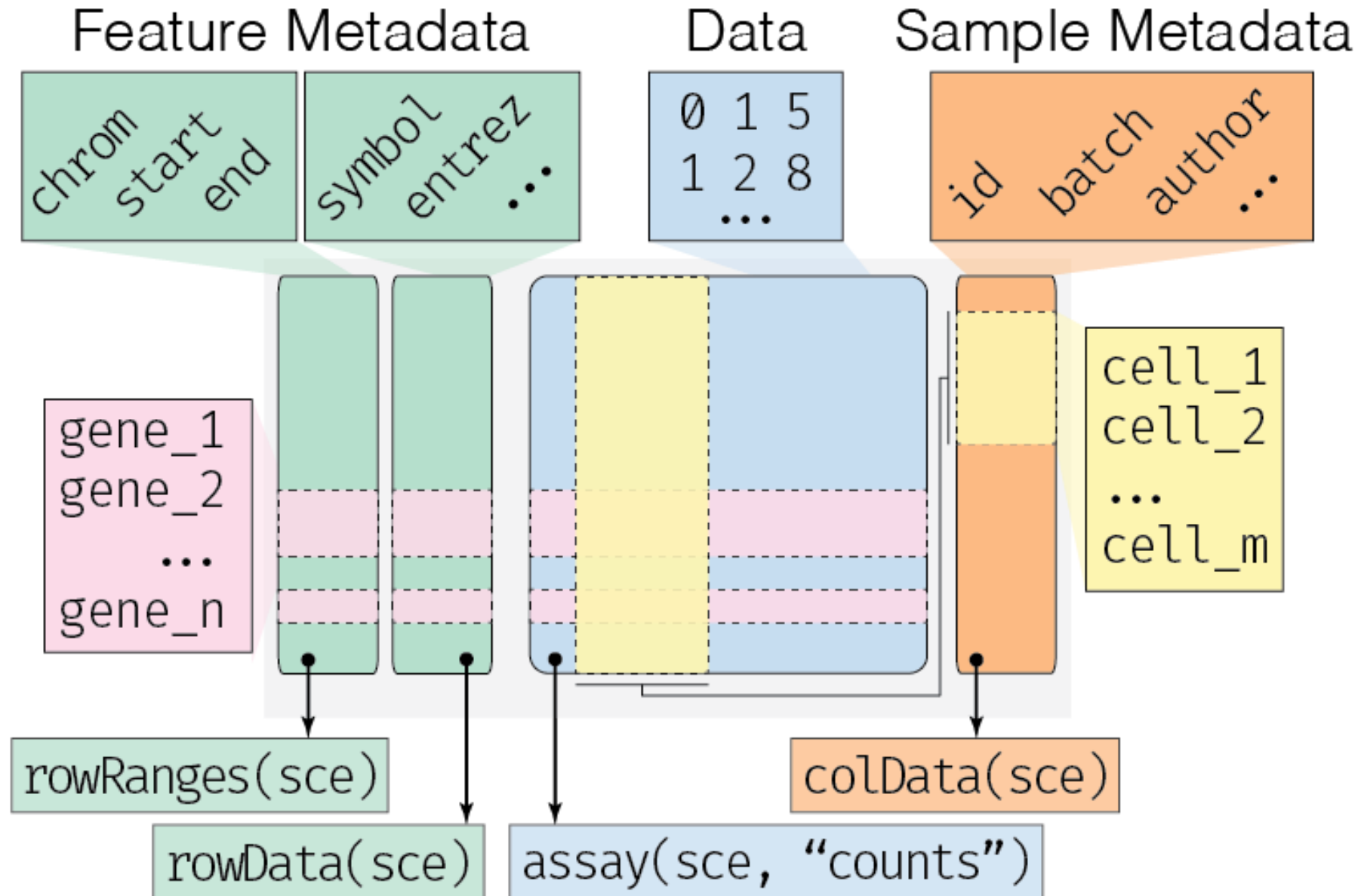
DAPI / morphology image (full resolution)

**Which framework have you used
for single-cell transcriptomics
data analysis?**

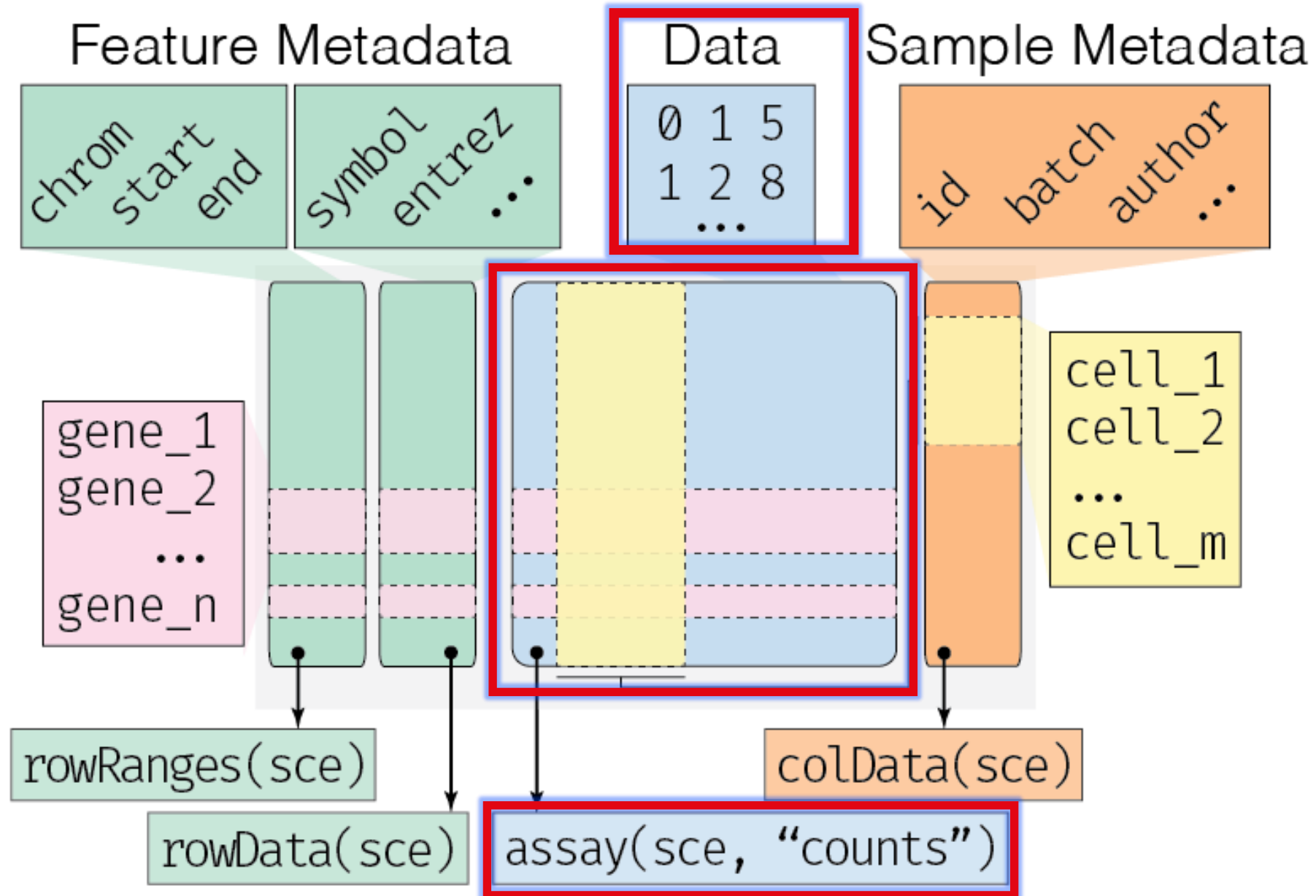
**Which framework have you used
for single-cell transcriptomics
data analysis?**

Seurat?

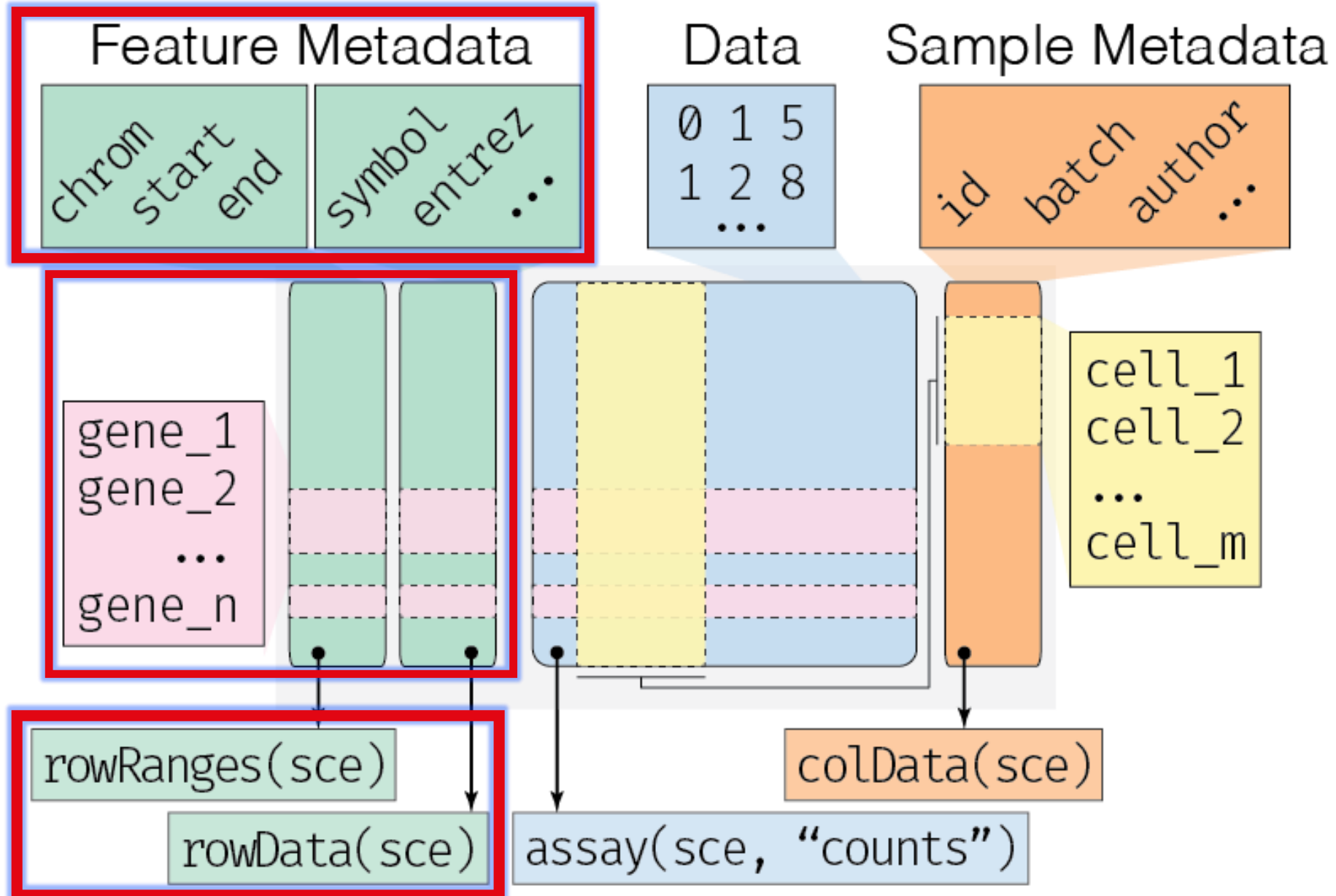
SingleCellExperiment (SCE)



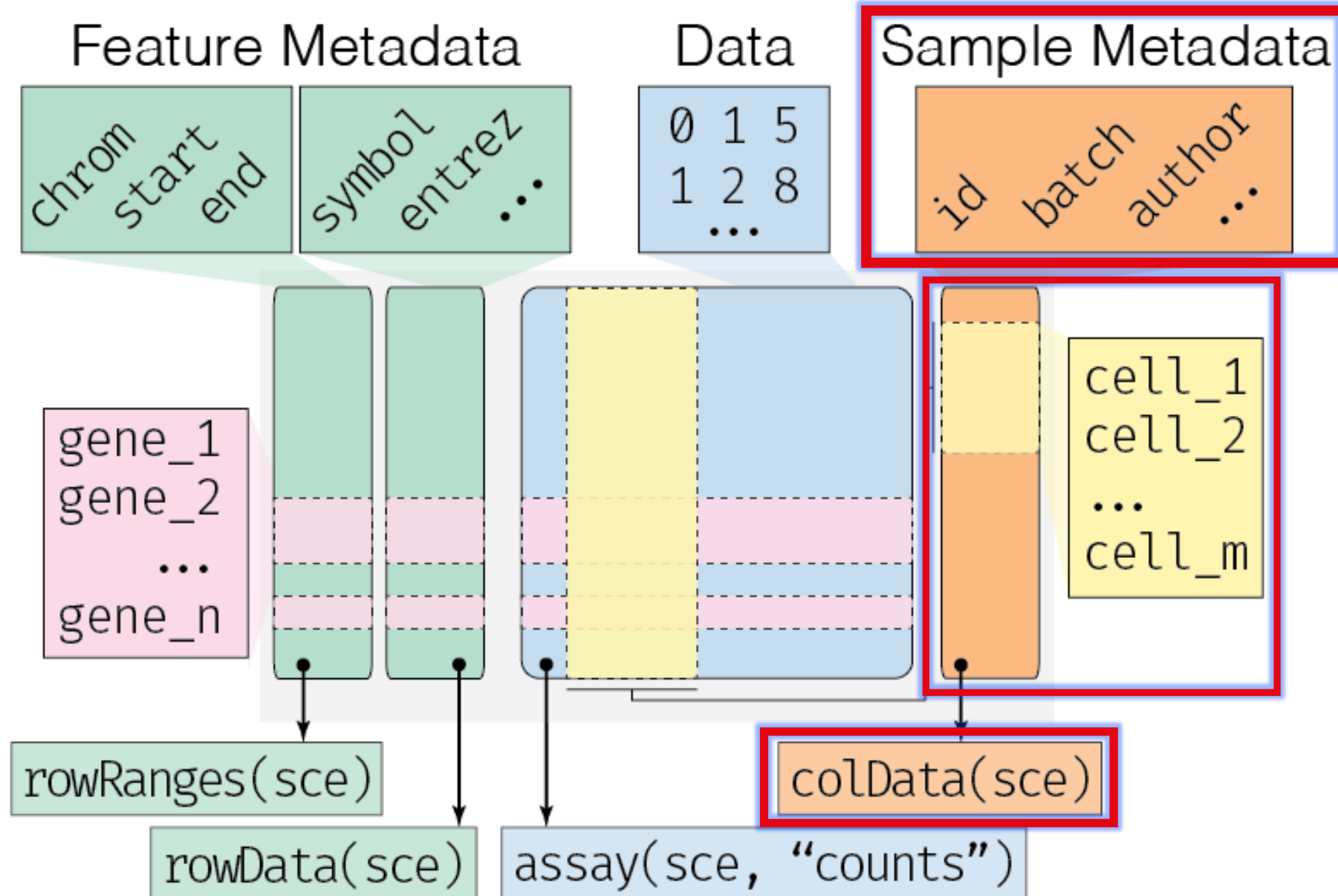
SingleCellExperiment



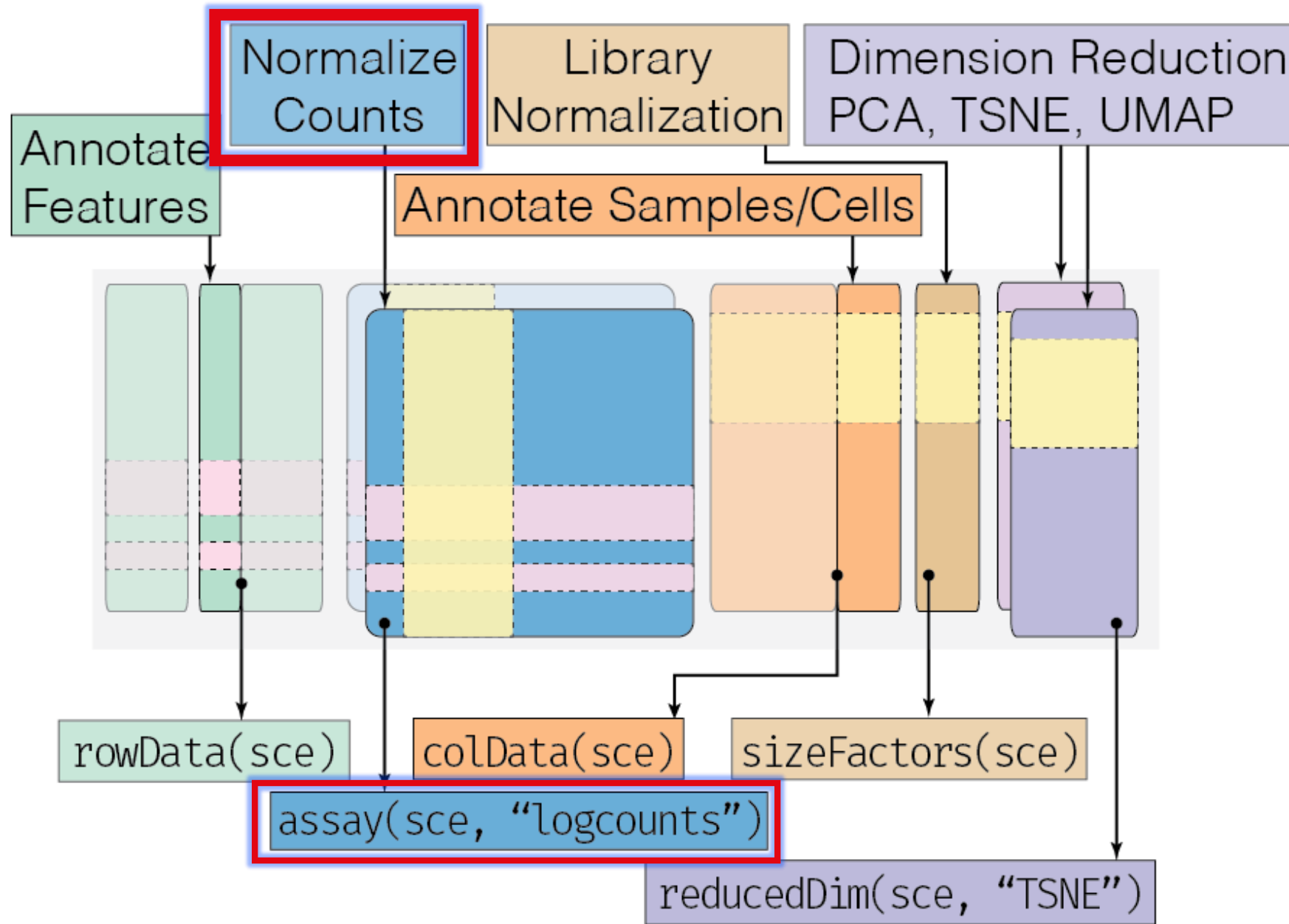
SingleCellExperiment



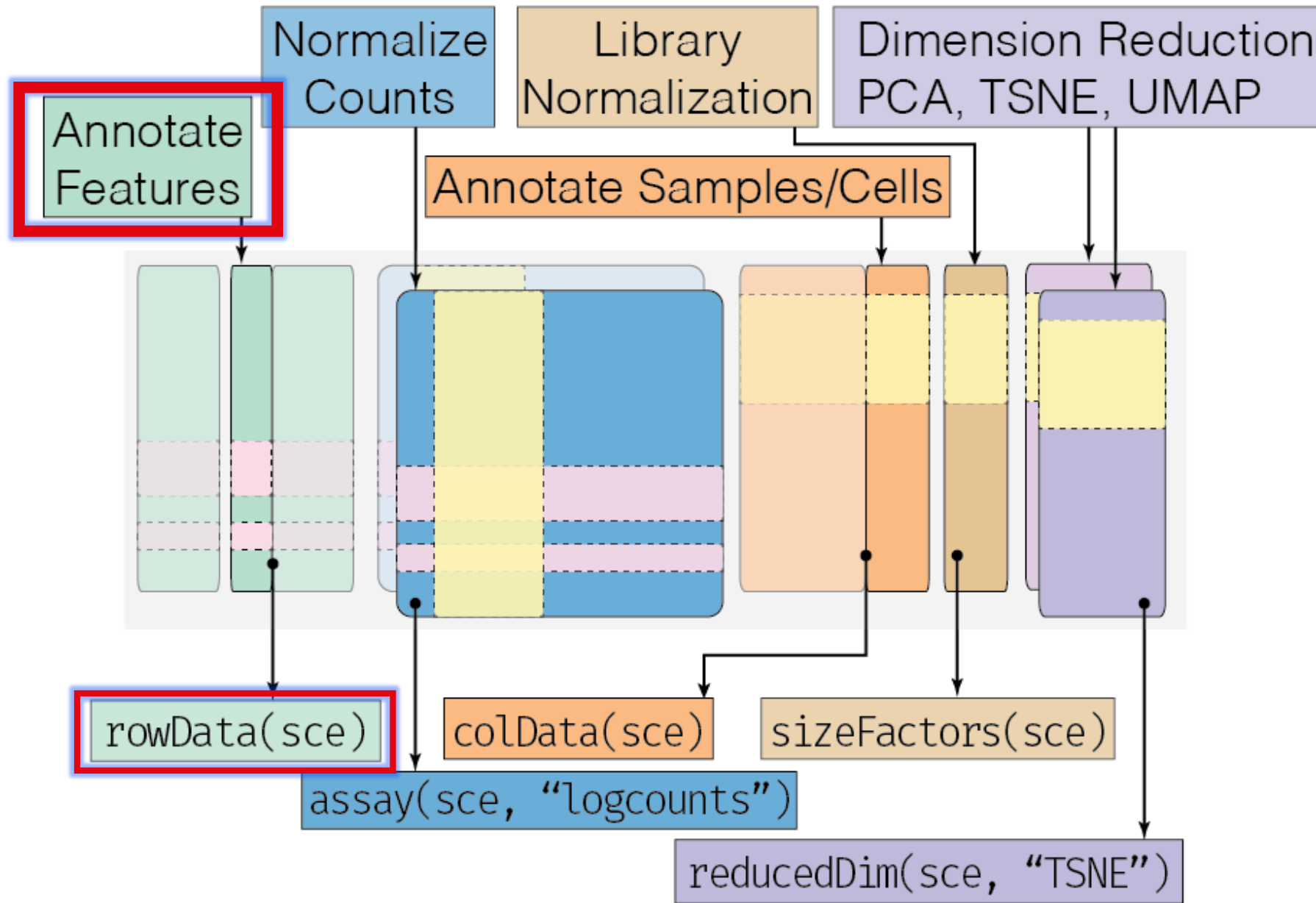
SingleCellExperiment



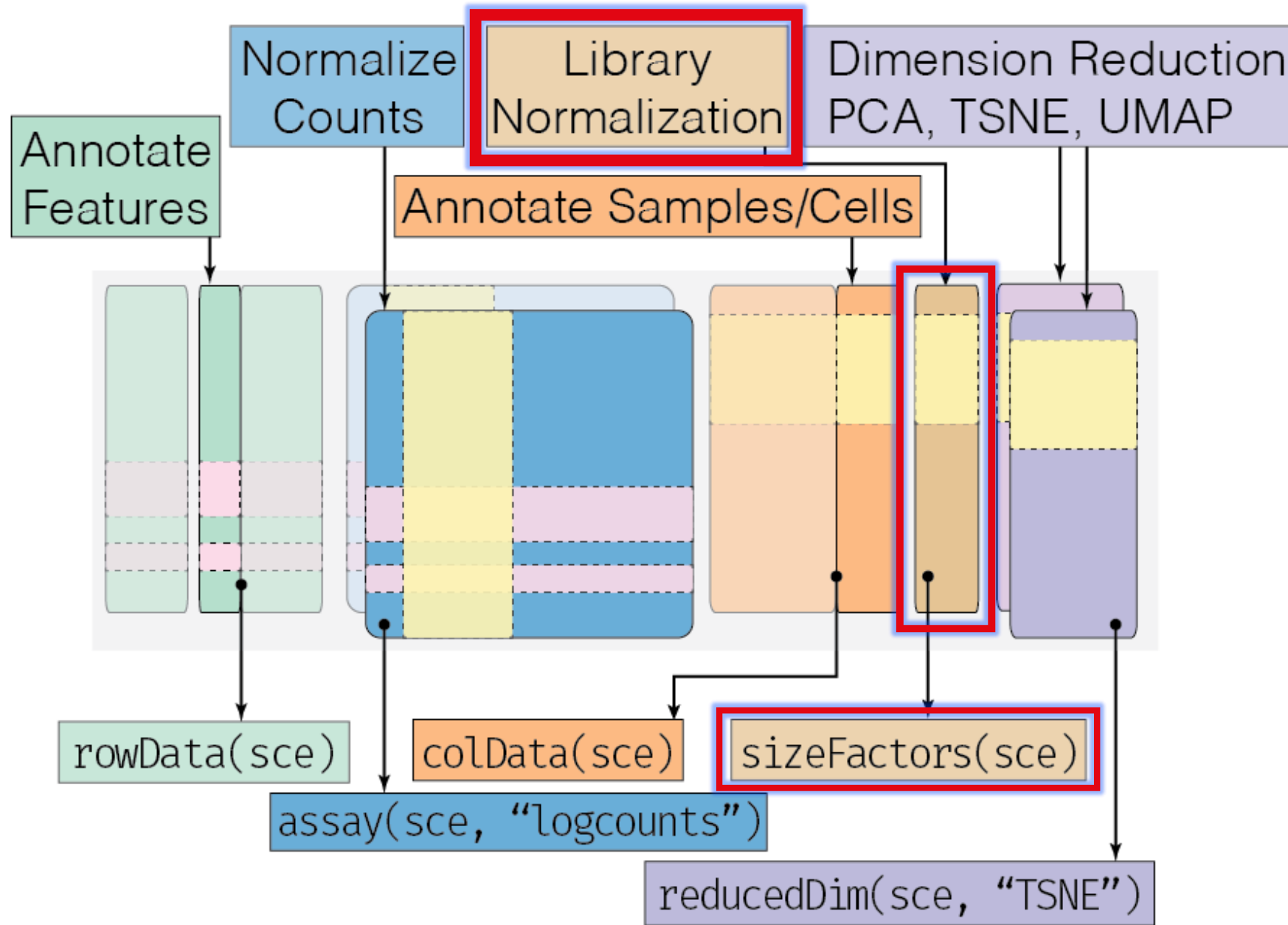
Extending SingleCellExperiment



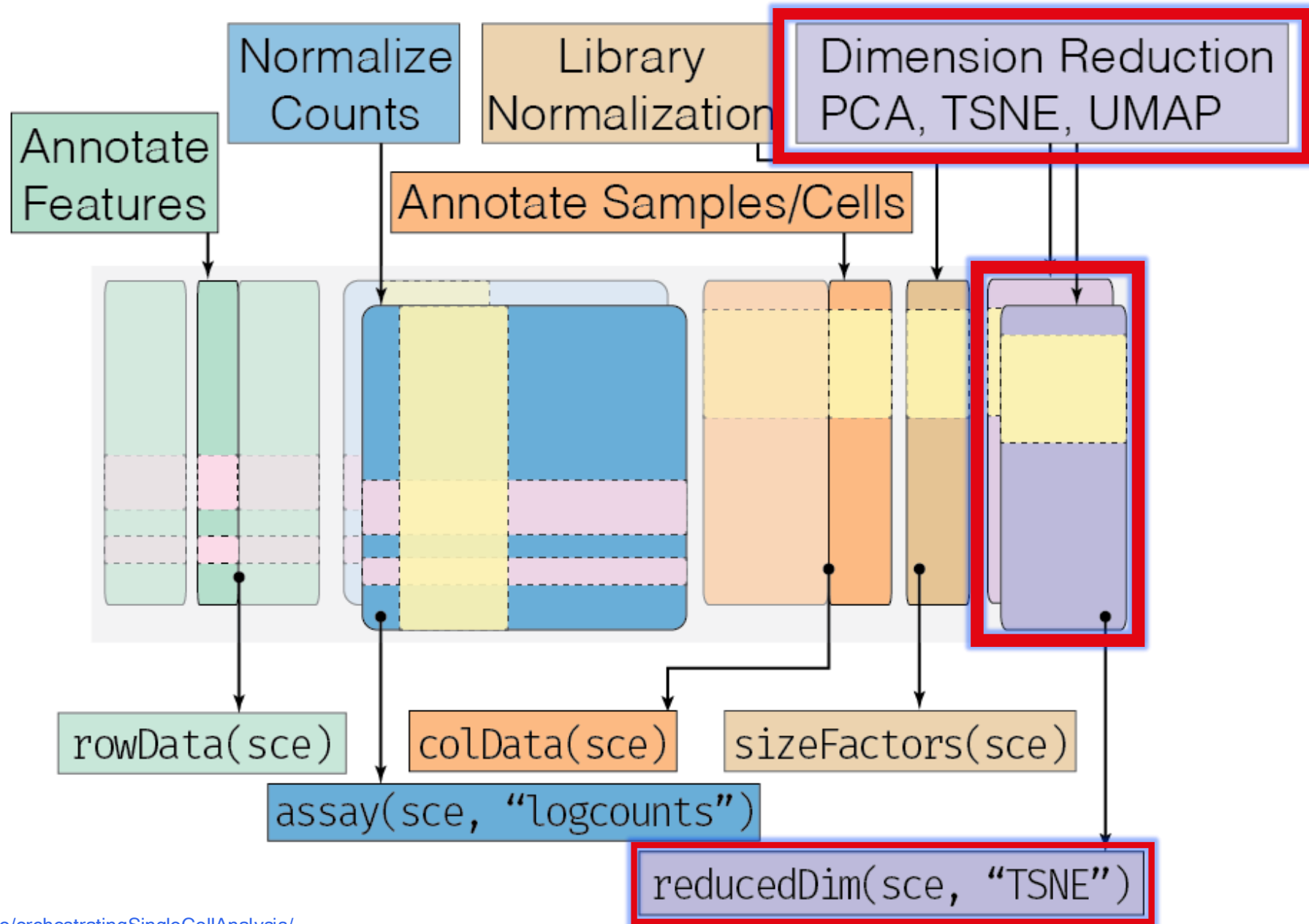
Extending SingleCellExperiment



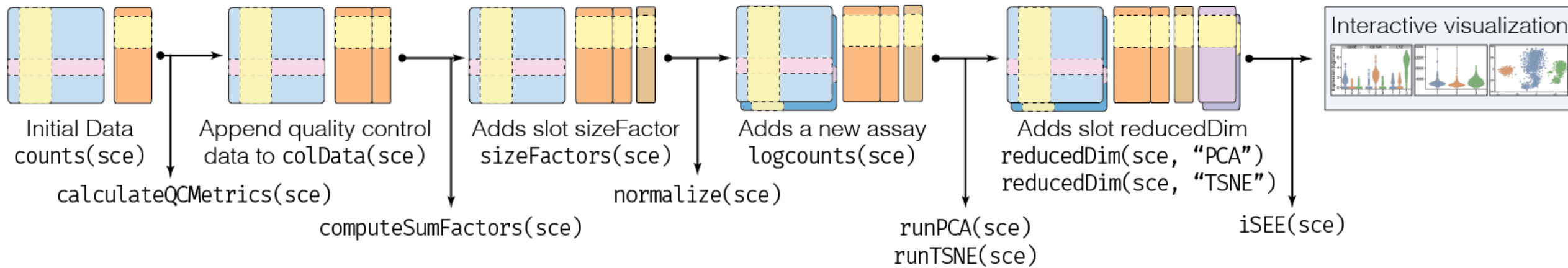
Extending SingleCellExperiment



Extending SingleCellExperiment



Example workflow of SingleCellExperiment extension



Can we use
SingleCellExperiment
for spatial transcriptomics data?

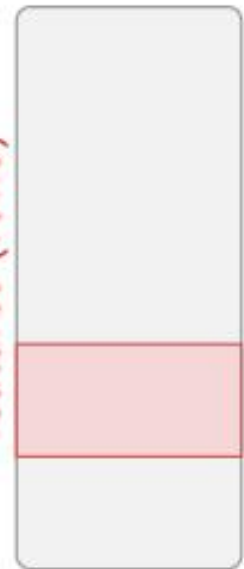
SpatialExperiment (SPE)

SpatialExperiment

SingleCellExperiment

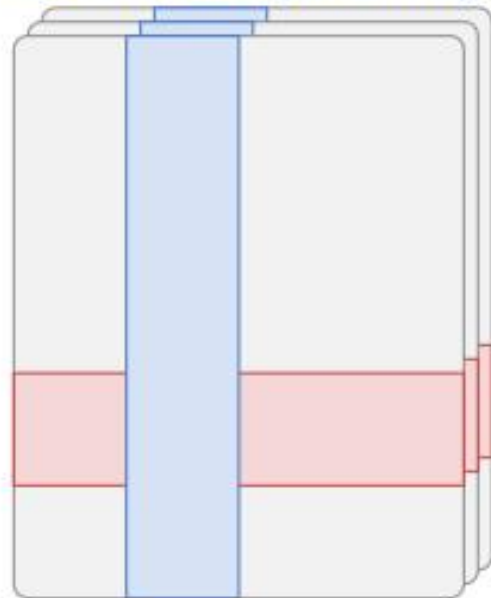
observations (columns)

features (rows)



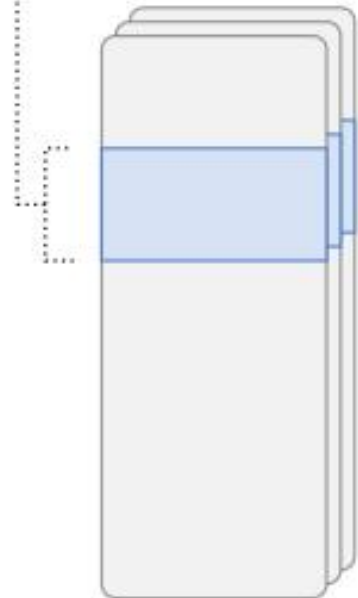
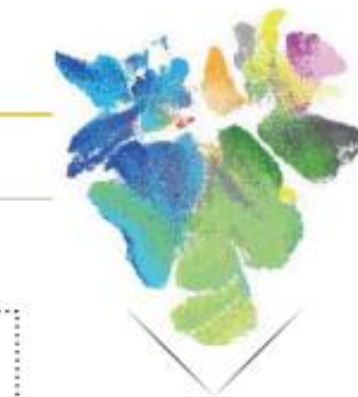
rowData

feature
metadata



assays

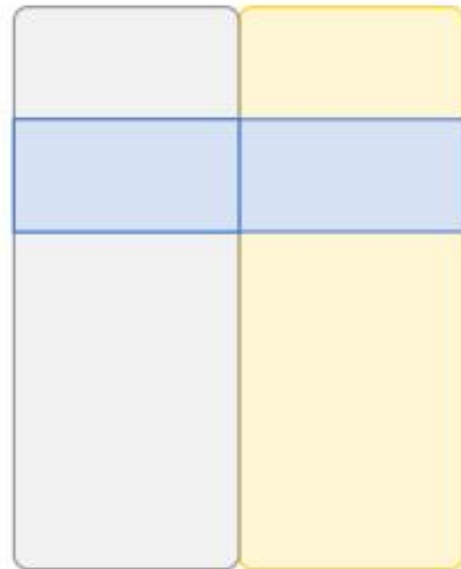
primary and
transformed data



reducedDims

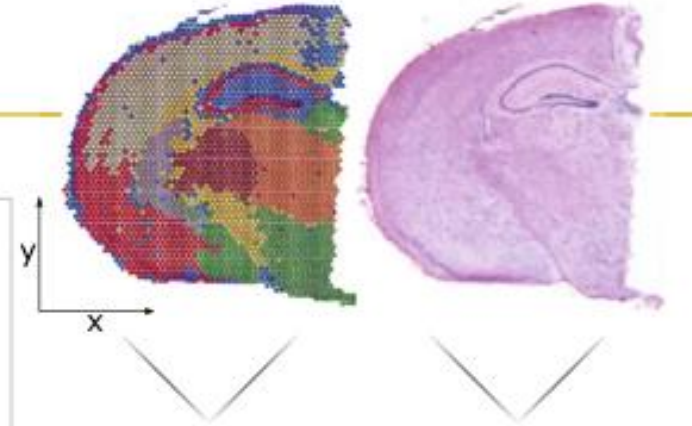
reduced
dimensions

sample id
condition
cell_type
...
barcode
in tissue
array_row
...



colData

observation
metadata spatial
metadata



spatialCoords

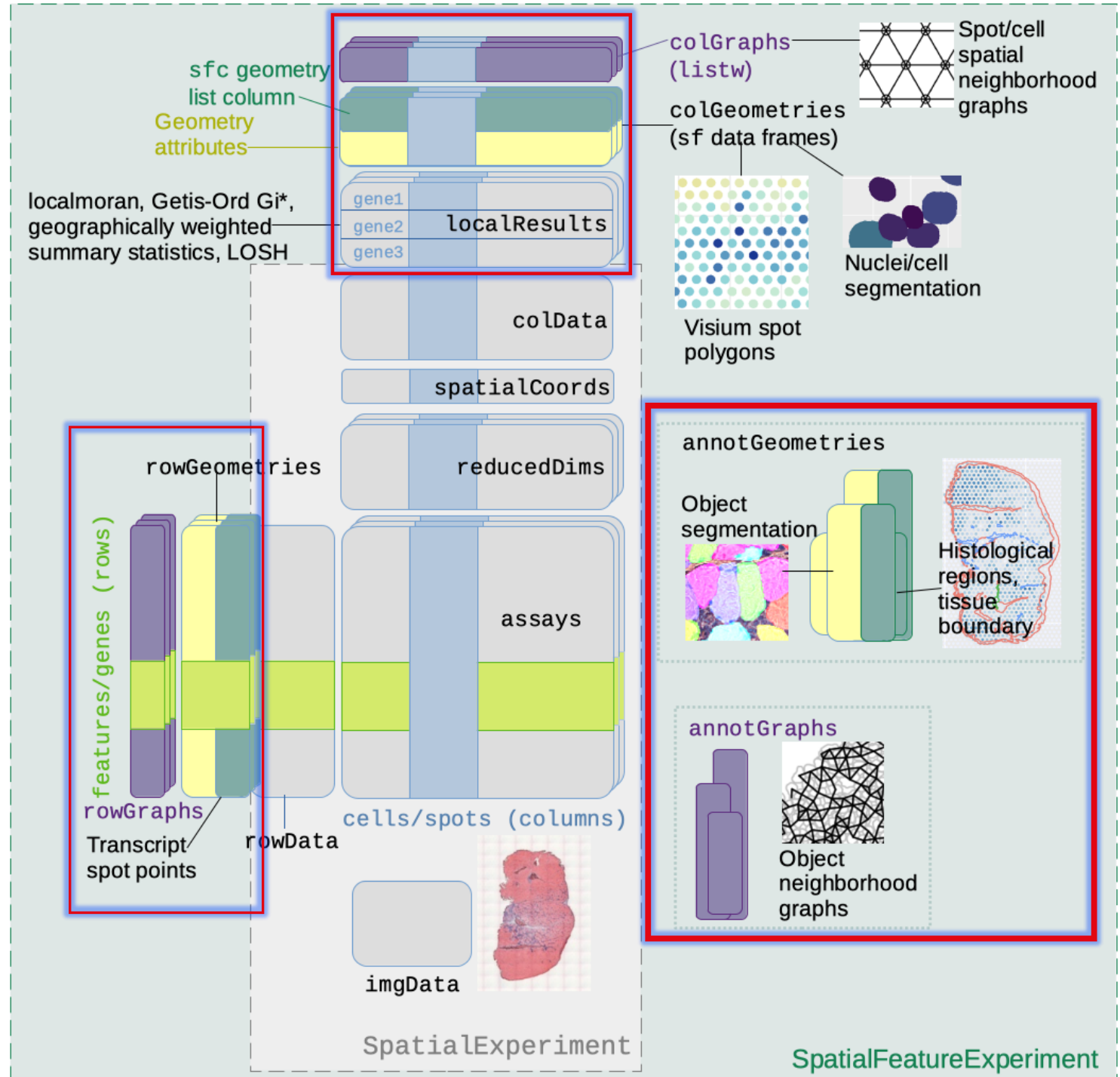
spatial
coordinates



imgData

image
metadata

SpatialFeature Experiment (SFE)



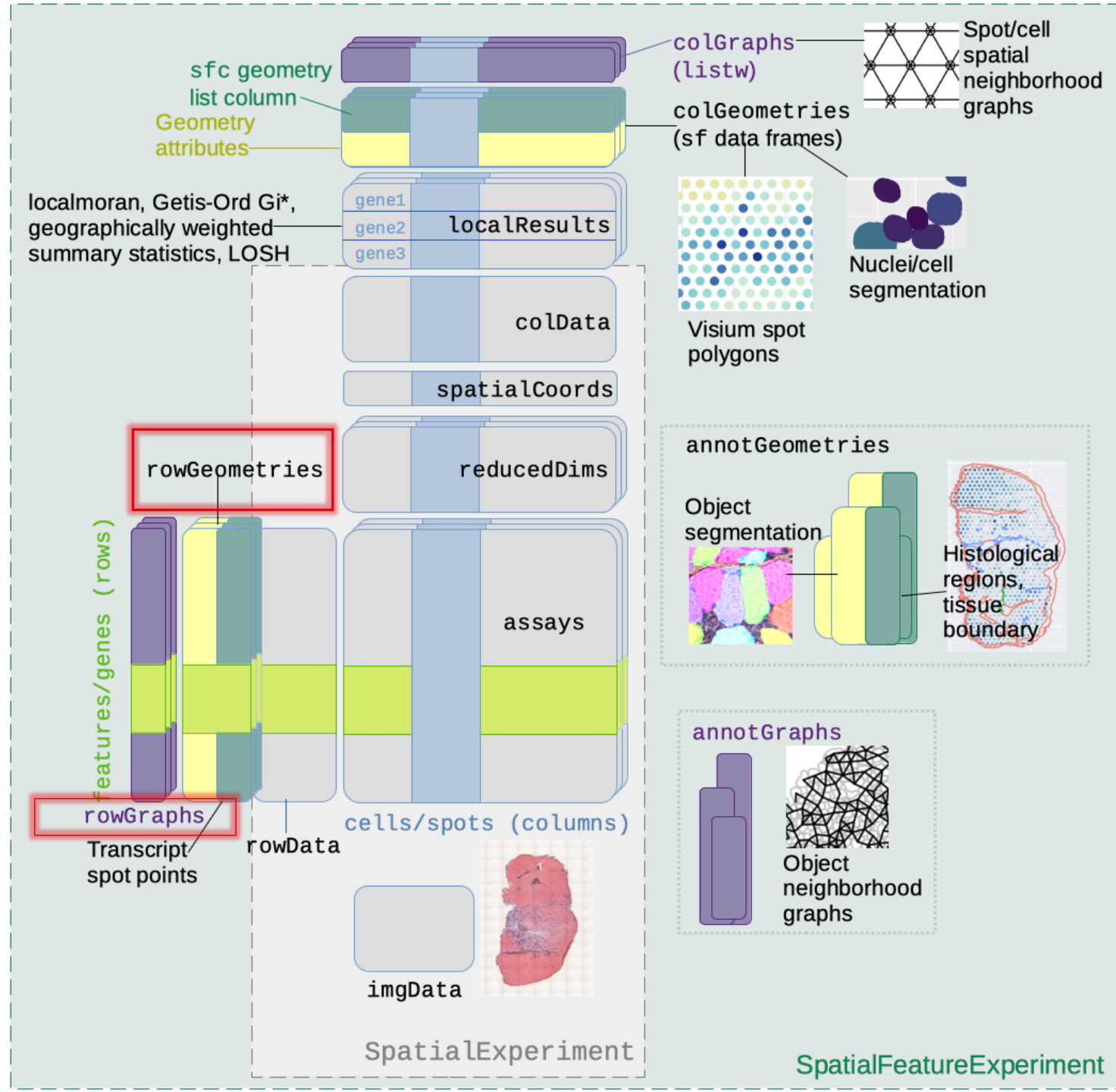
SpatialFeatureExperiment

rowGeometries: transcript location points per gene

The exact spatial coordinates (X, Y) of individual detected RNA molecules.

rowGraphs: transcript spatial graphs

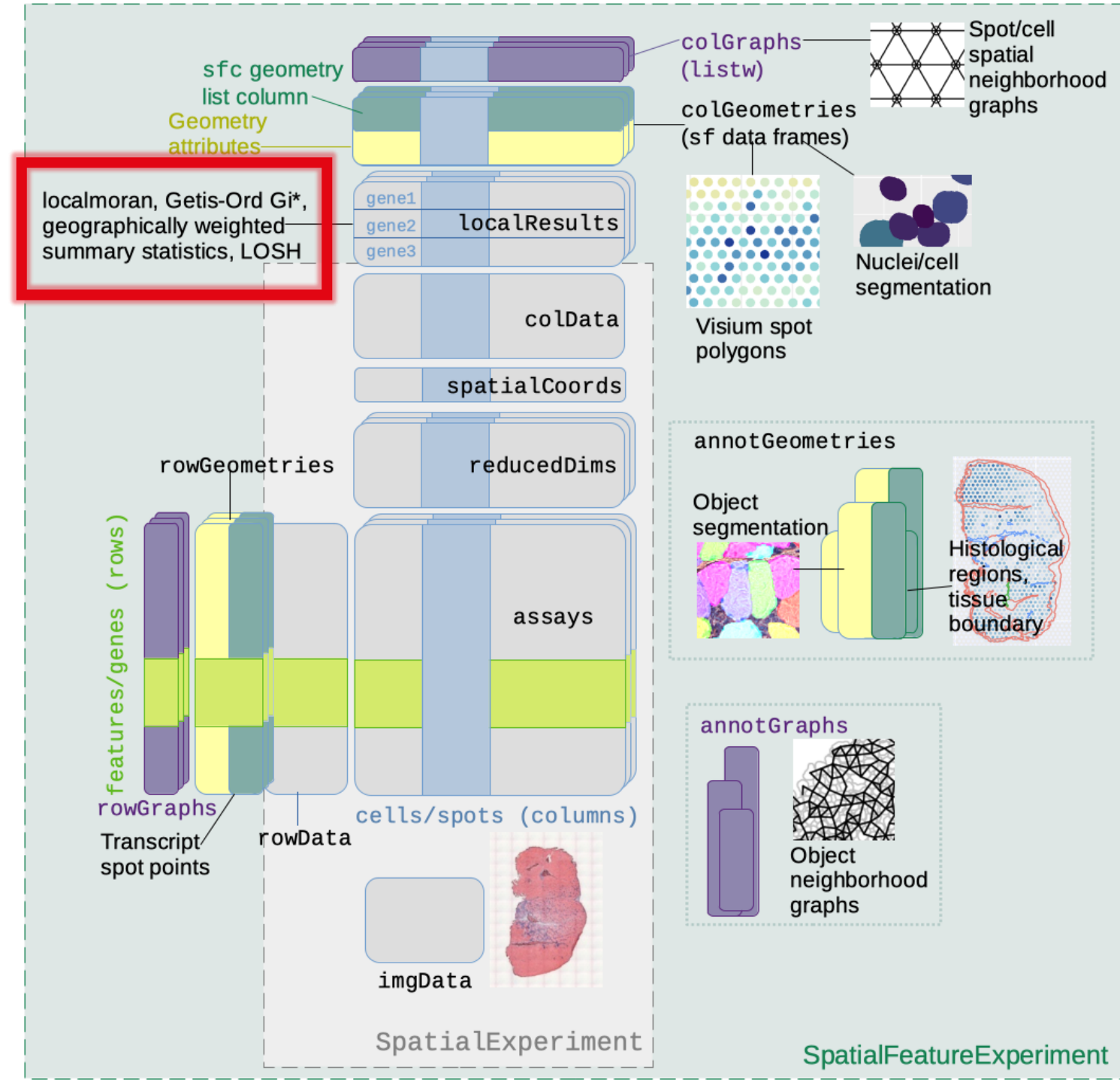
Network connections between individual transcript points to analyze molecule clustering.



SpatialFeatureExperiment

localResults: per-gene local spatial statistics (Moran's I, Getis-Ord Gi*, LOSH...)

Mathematical scores showing exactly where specific genes are significantly clustered or isolated.



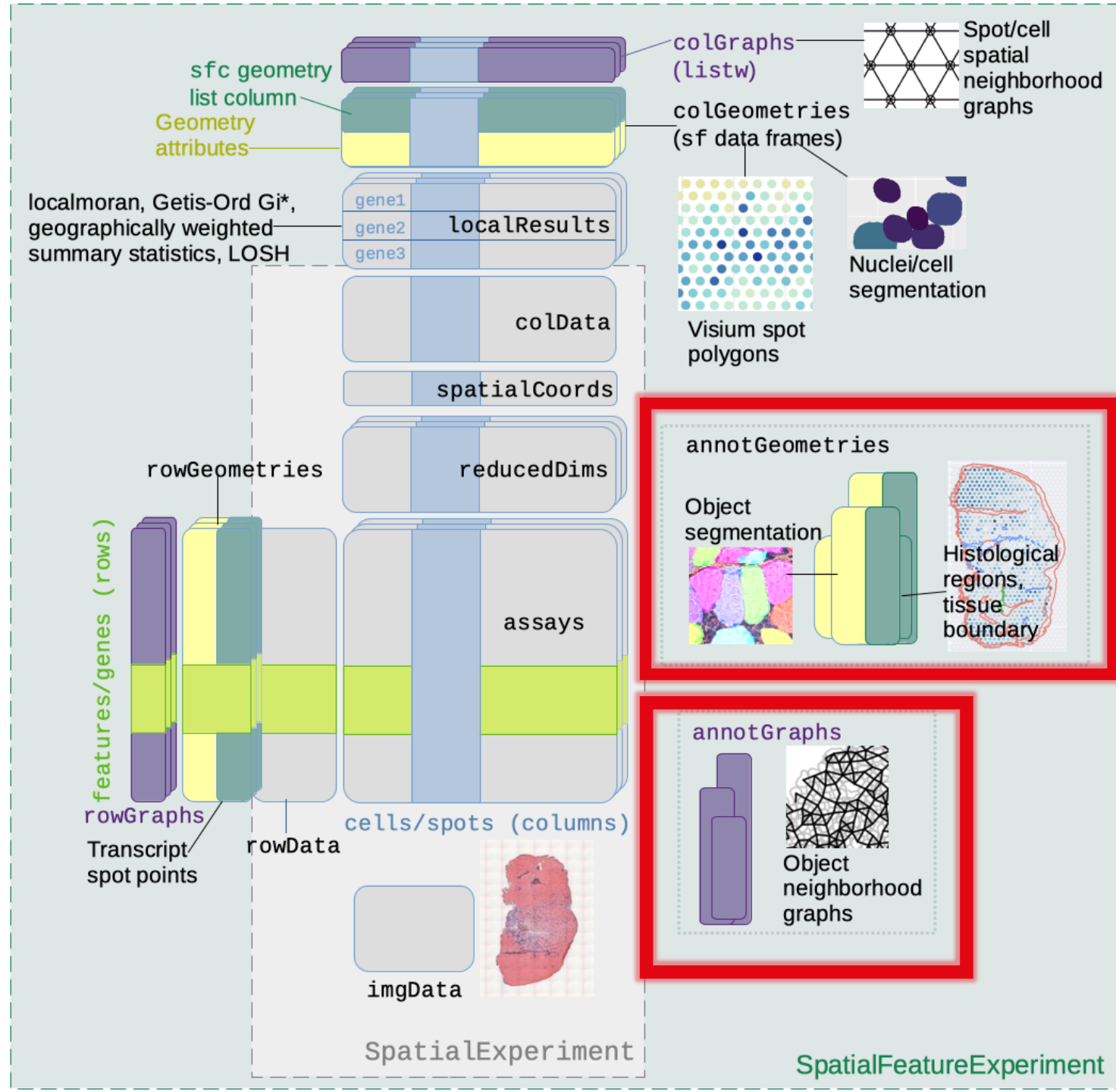
SpatialFeatureExperiment

annotGeometries: tissue regions, histological boundaries

Shapes outlining specific anatomical regions, such as tumor cores, cortex layers, or artifacts.

annotGraphs: object neighborhood graphs

Spatial networks showing how different tissue regions or macro-structures border one another.



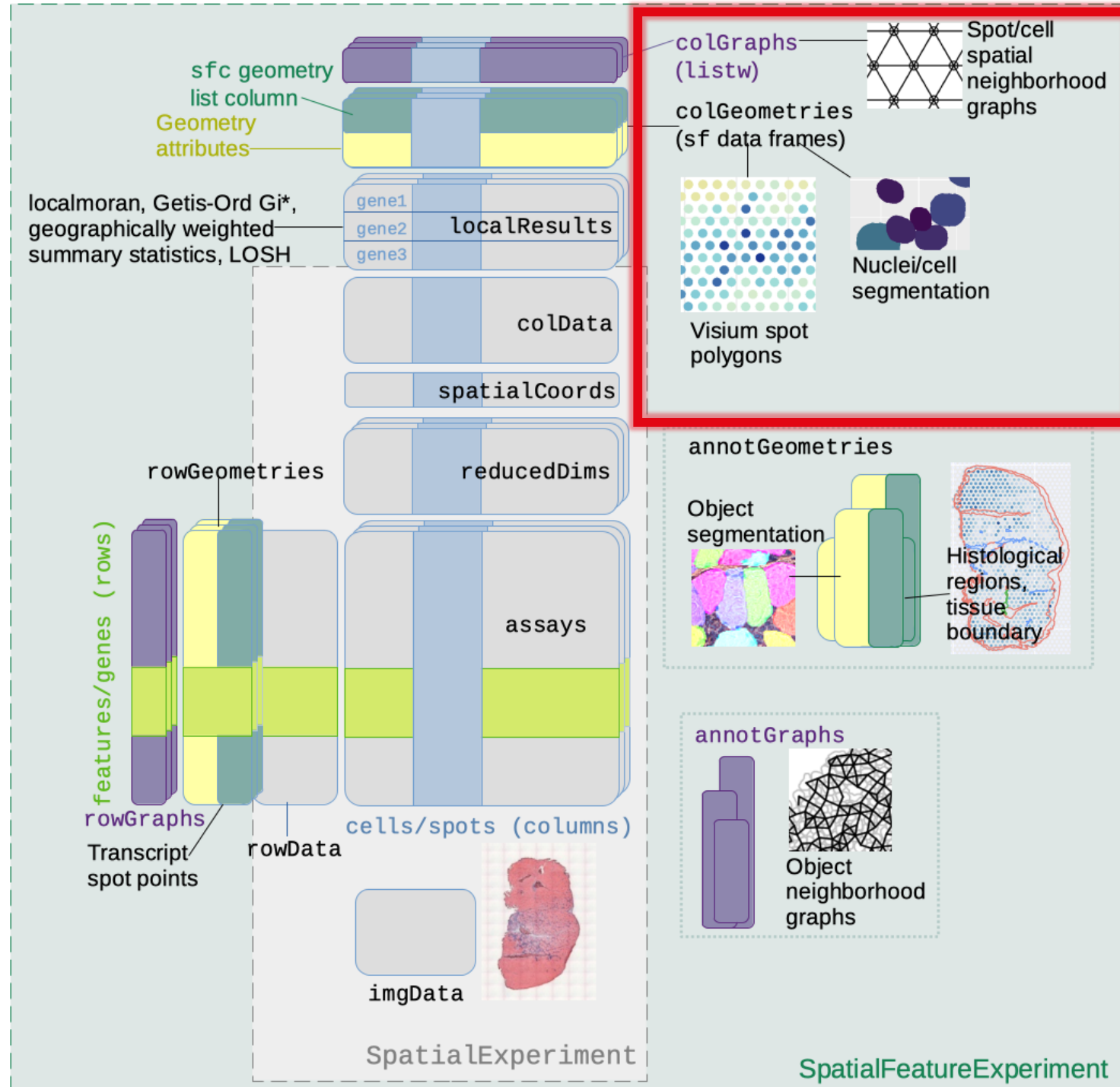
SpatialFeatureExperiment

colGeometries: cell/nucleus polygon shapes

The precise boundaries and shapes of individual cells or nuclei.

colGraphs: spatial neighborhood graphs

Networks mapping which cells (or bins) are physical neighbors.



SCE → SPE → SFE

Feature
Count matrix (assays)
Cell/spot metadata (colData)
Gene metadata (rowData)
Reduced dimensions (reducedDims)
Spatial coordinates (spatialCoords)
Tissue image (imgData)
Cell/nucleus polygons (colGeometries)
Transcript locations (rowGeometries)
Spatial neighbor graphs (colGraphs)
Tissue annotations (annotGeometries)
Typical use case

SCE	SPE	SFE
✓	✓	✓
✓	✓	✓
✓	✓	✓
✓	✓	✓
—	✓	✓
—	✓	✓
—	—	✓
—	—	✓
—	—	✓
—	—	✓
scRNA-seq	Visium HD	Xenium

Which framework will you use for
single-cell data?

Which framework will you use for
single-cell data?

SingleCellExperiment

Which framework will you use for
Visium HD with segmented
output data?

Which framework will you use for
Visium HD with segmented
output data?

SpatialFeatureExperiment

Which framework will you use for
Visium HD without segmented
output data?

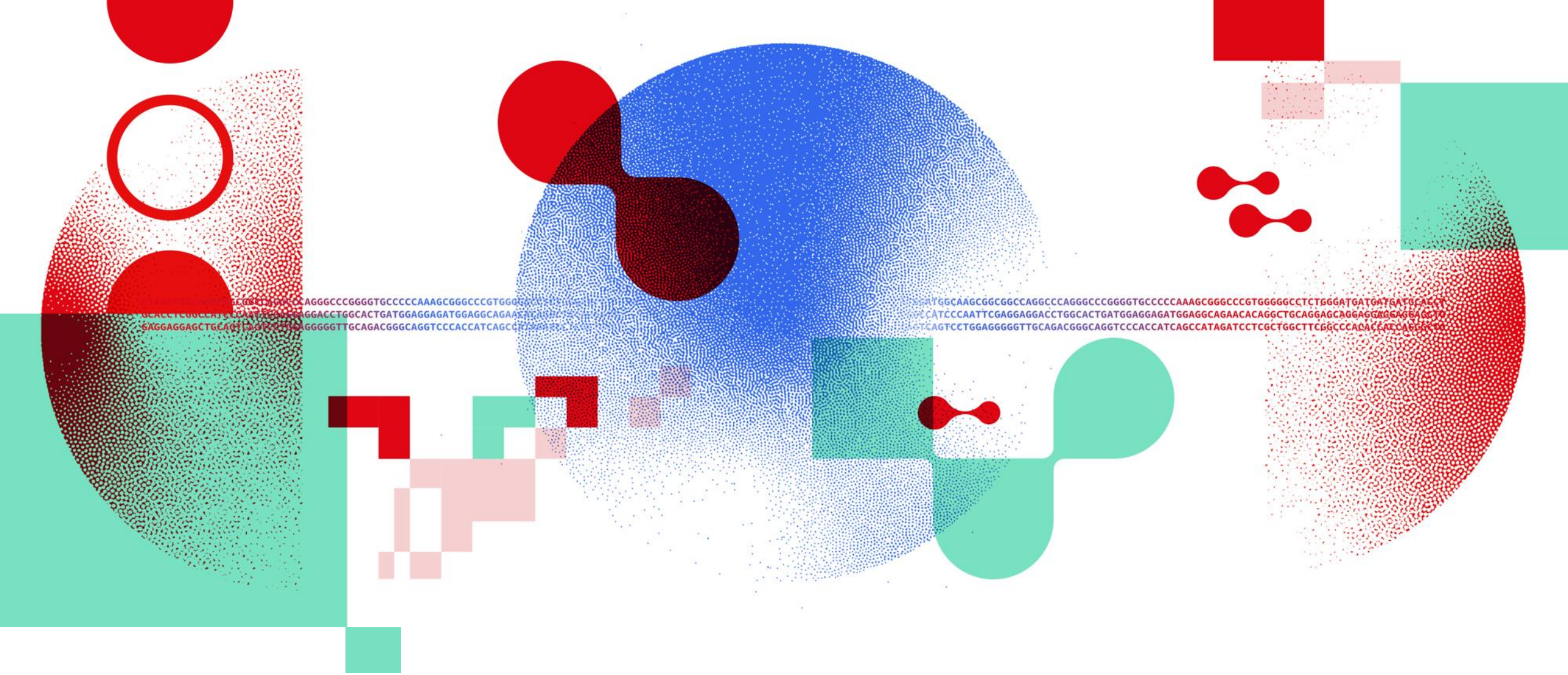
Which framework will you use for
Visium HD without segmented
output data?

SpatialExperiment

Which framework will you use for
Xenium data?

Which framework will you use for
Xenium data?

SpatialFeatureExperiment



Thank you

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